AN M-REGULAR COHERENT SHEAF ON PROJECTIVE SPACE IS GLOBALLY GENERATED AFTER TWISTING BY M

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Existence of a twist of a coherent sheaf on projective space that is globally generated. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-coherent-projective (tag 0AG6). Supplies the auxiliary degree d at which the twist is globally generated.
- Multiplication of sections of an m-regular sheaf. Stacks Project, Varieties, Lemma lemma-m-regular-multiply (tag 0ABP). Expresses sections of the higher twist through the base sections, which is what transfers generation downwards.

Source reference. Original reference 01YS: sources/stacks-chow/tags.txt, source lines 2567–2567. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08A7: sources/stacks-chow/tags.txt, source lines 10522–10522. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. If $\mathcal{F}$ is m-regular, then $\mathcal{F}(m)$ is globally generated.*

Proof. For all $d \gg 0$ the sheaf $\mathcal{F}(d)$ is globally generated. This follows for example from the first part of Cohomology of Schemes, Lemma 01YS. Pick $d \geq m$ such that $\mathcal{F}(d)$ is globally generated. Choose a basis $f_1, \ldots, f_r \in H^0(\mathbf{P}^n_k, \mathcal{F})$. By Lemma 08A7 every element $f \in H^0(\mathbf{P}^n_k, \mathcal{F}(d))$ can be written as $f = \sum P_i f_i$ for some $P_i \in k[T_0, \ldots, T_n]$ homogeneous of degree $d - m$. Since the sections f generate $\mathcal{F}(d)$ it follows that the sections f_i generate $\mathcal{F}(m)$. □

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